

Single-Cell Foundation Models for Cell Type Annotation and Perturbation Prediction: A Comprehensive Survey

Abstract

The advent of single-cell RNA sequencing (scRNA-seq) has fundamentally transformed the study of cellular heterogeneity, offering unprecedented resolution into the transcriptomic landscapes of individual cells across diverse tissues, developmental stages, and disease states. Historically, computational workflows have relied on highly variable gene selection, dimensionality reduction techniques, and task-specific algorithms to decode these complex datasets. Recently, however, inspired by the profound success of large language models in natural language processing, the computational biology field has witnessed a massive paradigm shift toward the development and deployment of single-cell foundation models (scFMs). Trained via self-supervised learning objectives on tens to hundreds of millions of transcriptomes, these deep learning architectures aim to learn universal, transferable representations of cellular states that can generalize across disparate biological contexts. This comprehensive survey critically examines the architectural designs, training methodologies, and downstream applications of single-cell foundation models, with a concentrated focus on two critical analytical tasks: cell type annotation and perturbation response prediction. By systematically comparing foundational models such as scGPT, Geneformer, scFoundation, and state-space architectures including scMamba and SC-MAMBA2, this report elucidates the diverse tokenization strategies, gene encoding mechanisms, and pretraining objectives that define the current technological landscape. Furthermore, the analysis evaluates the empirical performance of these large-scale models against established classical machine learning baselines—such as scVI, Harmony, CPA, and GEARS—through the lens of standardized, large-scale benchmarking frameworks like CellBench-LS and PertEval-scFM. Ultimately, this survey highlights the inherent tension between the theoretical promise of zero-shot model generalization and the practical reality of out-of-distribution brittleness, proposing critical future directions for the development of robust, multimodal, and biologically interpretable AI-driven virtual cells.

Introduction

Single-cell RNA sequencing (scRNA-seq) has emerged as a cornerstone of contemporary biological research, enabling the precise quantification of gene expression across millions of individual cells and facilitating a deeper understanding of cellular ontology, tumor microenvironments, and developmental trajectories¹. The resulting transcriptomic datasets are inherently high-dimensional, encompassing tens of thousands of genes, while simultaneously suffering from extreme sparsity (high dropout rates) and heavy susceptibility to technical noise

and batch effects introduced during the sequencing process². Traditional computational pipelines for scRNA-seq analysis involve highly engineered, stepwise procedures—typically beginning with rigorous quality control and the selection of highly variable genes (HVGs), followed by linear or non-linear dimensionality reduction (e.g., PCA, UMAP, t-SNE), batch correction using specialized tools like Harmony or Seurat, and concluding with clustering via Louvain or Leiden graph-based community detection algorithms⁴. While highly effective and widely adopted, these domain-specific models often require bespoke, iterative parameter tuning for new datasets, limiting their capacity for rapid, out-of-the-box generalization across diverse biological tissues, disease states, and divergent species⁷.

The introduction of the Transformer architecture and the broader foundation model paradigm has offered a highly compelling alternative to classical pipelines⁸. Foundation models operate on the premise that massive-scale, self-supervised pretraining on highly diverse, unlabeled datasets allows a model to learn universal latent structures and intrinsic data manifolds¹⁰. Once these general-purpose representations are learned, the model can be efficiently fine-tuned or probed for a multitude of specific downstream tasks using minimal labeled data¹⁰. In the context of computational biology, this linguistic analogy translates genes into discrete "words" and individual cells into "sentences," paving the way for transformer-based models that ingest raw or minimally processed count matrices to produce dense, contextualized cell embeddings¹.

Over the past three years, a wave of single-cell foundation models—including scBERT⁹, scGPT¹³, Geneformer¹⁷, scFoundation², Universal Cell Embeddings (UCE)¹⁹, and scPRINT²¹—have been trained on vast corpora ranging from 30 million to over 350 million distinct cells. These models aim to condense the vast combinatorial space of gene regulation into mathematical representations that inherently understand biological state, transitioning the field from descriptive analytics toward predictive and generative biology²³.

Despite the rapid proliferation and widespread adoption of these models, profound questions remain regarding their true efficacy, optimal scaling laws, and the architectural inductive biases required to effectively model complex biological systems. While scFMs exhibit exceptional performance on well-represented tasks such as intra-dataset cell type annotation and basic batch integration, their capability to generalize to complex, out-of-distribution (OOD) scenarios—particularly in mechanistically predicting the transcriptional outcomes of novel genetic or chemical perturbations—remains heavily contested in recent literature²⁵.

Independent, large-scale benchmarking initiatives have repeatedly demonstrated that simple, classical machine learning baselines often match or even outperform multi-million-parameter Transformers in label-scarce or perturbation-specific environments, highlighting a "baseline paradox" that challenges the necessity of massive over-parameterization⁵.

This survey aims to synthesize the current state of single-cell foundation models. The following sections detail the methodological foundations of these architectures, critically evaluate their deployment in cell type annotation and structural integration, and analyze their capacity for perturbation response prediction. By charting the transition from standard attention-based Transformers to linear-time state-space models, and from in-distribution memorization to genuine causal inference, this report outlines the trajectory of artificial intelligence toward the

realization of robust predictive virtual cells.

Methods: Architectures, Tokenization, and Pretraining

The construction of a single-cell foundation model requires resolving fundamental disparities between natural language text and biological omics data. Unlike text, where token sequence dictates grammar and meaning, gene expression profiles lack a sequential order; genes are expressed simultaneously within a cell, constituting an unordered set⁹. Furthermore, expression values are quantitative molecular counts rather than discrete categorical tokens².

Consequently, the design of scFMs diverges significantly from traditional large language models, necessitating unique approaches to tokenization, sequence modeling, and the formulation of self-supervised learning objectives⁸.

Tokenization and Gene Encoding Strategies

Tokenization in single-cell biology dictates how a cell's continuous or discrete read-count vector is mapped into an embedding space suitable for neural network processing. The choice of discretization strategy is a critical hyperparameter that profoundly impacts the model's sensitivity to technical noise, batch effects, and subtle but biologically significant signals³. The landscape of single-cell tokenization can be broadly categorized into several distinct paradigms.

Tokenization Strategy	Description	Representative Models	Strengths	Limitations
Bin-based Discretization	Expression values are partitioned into predefined, discrete numerical bins.	scBERT, scGPT, scMulan	Standardizes input distributions; mitigates the impact of extreme sequencing outliers.	Loss of fine-grained quantitative resolution; highly sensitive to bin hyperparameter selection ²⁹ .
Rank-based Discretization	Genes are sorted by expression level within the cell; absolute counts are discarded in favor of ordinal rank.	Geneformer, tGPT, LangCell	Highly robust to technical batch effects, varied sequencing depths, and platform-specific noise ²⁹ .	Sacrifices absolute magnitude data; struggles to differentiate cells with identical hierarchies but distinct total

				RNA volumes ²⁹ .
Value Projection (Continuous)	Log-normalized counts undergo non-linear transformation via multi-layer perceptrons (MLPs).	scFoundation, scPRINT	Preserves the full quantitative resolution of the data matrix ²² .	Diverges from traditional discrete NLP vocabularies, complicating purely autoregressive generation ²⁴ .
Patch-based Tokenization	Genomic regions or functional intervals are grouped into patches, treating regions as tokens.	scMamba	Bypasses highly variable feature selection; enables direct multi-omics integration (e.g., RNA + ATAC) ³⁰ .	Can obfuscate single-gene regulatory dynamics if patches are excessively broad.
Residual Vector Quantization (RVQ)	Compresses continuous expression profiles into discrete tokens drawn from ordered codebooks.	scPRINT-2, RVQ-Alpha	Maps continuous biology to discrete LLM vocabularies; tracks identity from broad lineages to functional states ²⁴ .	Requires complex teacher-student distillation frameworks to ground the discrete tokens to biological realities ²⁴ .

Bin-based discretization, utilized extensively by models such as scBERT and scGPT, offers a pragmatic compromise by grouping expression values into predefined bins, effectively standardizing the sequence modeling process while retaining relative expression intensities¹⁴. Each bin is assigned a learnable embedding, which is then combined with a unique gene identity embedding to form the final input token. However, this approach inherently flattens subtle transcriptional shifts if they fall within the boundaries of a single bin²⁹. Conversely, Geneformer relies on rank-based encoding, discarding absolute expression values entirely¹⁵. Genes are sorted by their relative expression level—often normalized against population-level frequencies to deprioritize ubiquitous housekeeping genes—and their

sequential rank serves as the primary informational signal¹⁵. This non-parametric approach confers remarkable robustness against technical batch effects but limits the model's ability to interpret overall transcriptional volume²⁹. Seeking to preserve absolute quantitative data, scFoundation and scPRINT utilize continuous value projection, directly embedding log-transformed counts through a dedicated MLP⁹. More recently, frameworks like scPRINT-2 have explored Residual Vector Quantization (RVQ), attempting to map continuous transcriptomic profiles into discrete tokens that perfectly align with the vocabularies of massive multi-modal LLMs, enabling autoregressive generation of cell states²⁴.

Gene Identity and Biological Priors

Beyond quantifying the expression value, foundation models must encode the identity and biological context of the gene itself. Early models in the space treated genes as arbitrary categorical tokens, learning their embeddings entirely from scratch during the self-supervised pretraining phase⁸. However, next-generation models increasingly inject structural and functional biological priors directly into the tokenization phase to accelerate convergence and improve generalization.

The Universal Cell Embedding (UCE) model and the scPRINT architecture leverage external protein language models—specifically the ESM-2 framework—to initialize gene embeddings¹⁹. By encoding genes using the amino acid sequence of their primary protein products, the foundation model inherits a vast reservoir of pre-computed evolutionary, functional, and structural biology knowledge²². This allows these models to generalize to unseen genes and perform cross-species analysis out-of-the-box, as orthologous genes across different species inherently possess highly similar structural embeddings in the ESM-2 latent space¹⁹.

Furthermore, models like scPRINT augment this with genomic positional encoding, allowing the model to intrinsically recognize that genes in close physical proximity on the chromosome are frequently co-regulated by shared cis-regulatory elements²². Alternatively, architectures like RegFormer explicitly bake Directed Acyclic Graphs (DAGs) of known Gene Regulatory Networks (GRNs) into the topological sorting of their inputs, directly hardcoding transcription factor-target relationships into the sequence structure³⁴.

Architectural Paradigms: From Transformers to State-Space Models

The dominant architectural paradigm in single-cell foundation modeling has been the Transformer, relying on multi-head self-attention mechanisms to capture the complex, many-to-many interactions characteristic of gene regulatory networks³. Models such as scGPT and Geneformer employ variations of the BERT (encoder-only) or GPT (decoder-only) frameworks⁸. However, the self-attention mechanism imposes a severe computational

bottleneck: memory and computational complexity scale quadratically ($O(n^2)$) with respect

to the sequence length n ²⁸. Given that the human genome contains over 20,000 protein-coding genes, standard Transformers are often forced to arbitrarily truncate their inputs, relying exclusively on a highly filtered subset of 2,000 to 10,000 Highly Variable Genes (HVGs) per cell³⁰. This necessary truncation invariably discards lowly expressed transcription

factors, non-coding RNAs, and subtle regulatory genes that are frequently critical for determining fine-grained cell states and mechanisms of disease³¹. To overcome this quadratic bottleneck, recent literature demonstrates a rapid and highly promising pivot toward State-Space Models (SSMs), specifically leveraging the Mamba architecture²⁸. Models such as GeneMamba, scMamba, and SC-MAMBA2 leverage state-space duality and bidirectional context aggregation to achieve linear time complexity ($O(n)$) while maintaining long-range contextual dependencies³⁷. SC-MAMBA2, boasting over 150 million parameters and trained on a corpus of 57 million cells, is currently capable of ingesting ultra-long transcriptome sequences encompassing over 60,000 genes without the need for arbitrary highly variable feature filtering³⁸. By processing the entire transcriptome simultaneously, these models can integrate signals from across the whole genome, bridging the gap between highly localized expression profiling and holistic systems biology³⁸. Similarly, scMamba introduces patch-based tokenization to state-space models, processing genomic regions linearly to enable the seamless integration of highly sparse multi-omics data (such as RNA paired with chromatin accessibility) while preserving essential genomic positional information³¹.

Pretraining Objectives and Latent Space Formation

The self-supervised objectives utilized to train these architectures ultimately dictate the topological structure and utility of their latent spaces. The most prevalent objective remains Masked Language Modeling (MLM), wherein a subset of a cell's expressed genes is masked, and the model must reconstruct their values using the surrounding unmasked gene context¹⁴. While effective at capturing local co-expression networks, MLM faces challenges in single-cell data due to extreme dropout rates. To counteract this, models like scPRINT and PerturbNet replace standard mean-squared error (MSE) reconstruction losses with Zero-Inflated Negative Binomial (ZINB) loss functions²¹. The ZINB distribution explicitly models the high probability of zero-counts inherent in single-cell sequencing technologies, leading to vastly superior expression denoising capabilities compared to standard loss functions²¹. Recent comprehensive evaluations highlight a significant paradox in model pretraining: standard MLM objectives can achieve extremely low pretraining loss without producing high-quality, transferable cell-level representations for downstream tasks⁴⁴. This occurs because models can easily guess missing gene values by relying on highly localized, redundant co-expression clusters without fundamentally compressing the global state of the cell into a unified embedding. To address this, the Whole-Cell Expression Decoding (WCED) objective has been introduced⁴⁴. Under WCED, the model is forced to decode the entire genome's expression vocabulary from a single constrained bottleneck (such as a [CLS] token embedding), rather than relying on unmasked neighboring tokens⁴⁴. This forces the architecture to learn maximally informative, highly compressed global cellular representations, which has been shown to significantly boost zero-shot generalization across batch integration and cell-type annotation tasks⁴⁴.

Methods: Cell Type Annotation and Integration

Cell type annotation—the precise assignment of a biological identity to a single-cell transcriptome—is the foundational requisite task of all scRNA-seq analysis pipelines³. Traditional methods have historically relied on unsupervised clustering followed by the manual, often subjective inspection of canonical marker genes³. Reference-based annotation tools improved upon this but remained severely limited by the size and quality of specific reference atlases. Single-cell foundation models promise to automate and objectify this entire pipeline through data-driven embeddings that group cells based on their intrinsic, global transcriptomic manifolds, effectively bypassing the need for manual marker curation.

Zero-Shot Annotation and Universal Embeddings

A primary value proposition of scFMs is their ability to perform high-fidelity zero-shot annotation—classifying cells directly from their embeddings without requiring any task-specific fine-tuning or retraining on labeled target datasets⁶. Because foundation models are exposed to millions of diverse cellular states during pretraining, their high-dimensional latent spaces naturally organize cells by biological lineage and developmental hierarchy¹⁹.

The Universal Cell Embedding (UCE) model exemplifies this powerful zero-shot capability. Pretrained on an Integrated Mega-scale Atlas of 36 million cells spanning dozens of tissues and eight species, UCE projects diverse transcriptomes into a unified biological latent space¹⁹. Experimental results forcefully demonstrate that UCE exhibits emergent behavior, generating biologically accurate embeddings for cells from entirely novel, non-model species (such as the naked mole-rat, green monkey, and chicken) that were completely absent from its training corpus³³. By utilizing ESM-2 protein embeddings to represent genes, UCE intrinsically maps homologous biological functions into the same mathematical region. Consequently, rich reference labels from well-characterized human or mouse atlases can be directly transferred to novel species in a zero-shot manner via simple distance metrics (e.g., k-Nearest Neighbors or logistic regression applied directly to the latent space)²².

In parallel, Text-bridge Large Language Models (LLMs) offer a completely distinct annotation paradigm. Frameworks such as Cell2Sentence, CHATCELL, and scAgent translate sparse, high-dimensional gene expression profiles directly into natural language sequences (e.g., generating text such as: "This cell exhibits high expression of CD8A, GZMB, and IFNG, while lacking CD4")¹⁴. These synthesized textual sequences are subsequently fed into domain-adapted LLMs (like GPT-4 or specific biomedical agentic frameworks) that leverage their vast internal biomedical literature comprehension to output a semantic cell type label⁷. By explicitly bridging raw transcriptomics with the global corpus of scientific literature, these models overcome the "black-box" nature of pure numerical embeddings, providing interpretable reasoning chains, biological context, and ontology alignments for their annotations³².

Fine-Tuning and Batch Correction

While zero-shot embeddings are highly effective at revealing underlying biological structures

and broad lineages, they frequently fail to fully untangle complex technical batch effects—systematic discrepancies arising from different sequencing platforms (e.g., 10x Genomics vs. Smart-seq2), donor genetics, varying sequencing depths, or disparate laboratory protocols⁶. A systematic evaluation of zero-shot scFMs demonstrated that pronounced batch-specific information often remains heavily encoded in the pretrained embeddings, occasionally resulting in massive foundation models underperforming simple classical linear baselines like PCA or domain-specific tools like Harmony in data integration tasks⁵¹. To achieve optimal, atlas-scale data integration, scFMs must typically undergo targeted fine-tuning. Models like scGPT provide specialized architectural modules and contrastive learning objectives explicitly designed to project cells into a batch-invariant space¹⁶. During fine-tuning on a partially labeled dataset, the model dynamically adjusts its attention weights to prioritize biologically relevant expression modules while actively penalizing and suppressing batch-correlated noise vectors¹⁶. The comprehensive CellBench-LS benchmark indicates that while classical methods hold their ground in zero-shot tasks, fine-tuned foundation models consistently dominate in few-shot cell type classification and complex multi-batch harmonizations, successfully leveraging their pretrained priors to avoid catastrophic overfitting on limited target data⁶. Similarly, scPRINT-2, utilizing parameter-efficient fine-tuning (PEFT), has demonstrated state-of-the-art performance across both classification and batch correction on live community benchmarks, significantly outpacing earlier models²³.

Methods: Perturbation Response Prediction

Predicting how a specific cell will dynamically respond to an external intervention—such as a targeted CRISPR gene knockout, gene over-expression, or the introduction of a small-molecule therapeutic—is a critical frontier for drug discovery, precision medicine, and systems biology⁵². The ability to accurately simulate the transcriptomic shift from a basal (control) state to a perturbed state *in silico* offers the potential to entirely circumvent the prohibitive financial and temporal costs associated with physically screening vast, combinatorial intervention spaces *in vitro*⁵⁴.

The Combinatorial Challenge and Classical Baselines

The perturbation prediction task is defined mathematically as identifying a robust function that maps a control cell state matrix X_c to a predicted perturbed cell state X_p under a specified intervention condition. Early deep generative approaches, such as scGen and the Compositional Perturbation Autoencoder (CPA), conceptualized this challenge as a latent space vector arithmetic problem, drawing direct inspiration from word embedding manipulations in early NLP⁵⁶.

In CPA, a cell's overall expression profile is encoded into three independent, learnable additive embeddings corresponding to: the basal cellular state, the covariate state (e.g., cell line or tissue type), and the perturbation state (the specific drug or genetic intervention)⁵⁸. To simulate a novel condition, the model algebraically manipulates these vectors in the latent space (e.g.,

$Z_{predicted} = Z_{basal} - Z_{drug_A} + Z_{drug_B}$) which is then passed through a non-linear

decoder to reconstruct the expected post-perturbation gene matrix⁵⁷. While this modular approach is highly interpretable, allows for dose-response modeling, and is highly effective for predicting the effects of *seen* perturbations in *unseen* cellular contexts (an interpolation task), models like scGen and CPA structurally struggle with true extrapolation—predicting the system-wide response to a gene or drug that was entirely absent from the training set⁵².

Graph-Enhanced and Knowledge-Driven Extrapolation

To successfully address out-of-distribution (OOD) extrapolation to unseen genes, advanced models incorporate explicit biological prior knowledge. GEARS (Graph-Enhanced Gene Activation and Repression Simulator) currently represents the benchmark state-of-the-art for predicting genetic perturbations⁵⁵. Recognizing that genes do not act in isolation, GEARS integrates single-cell expression data with extensive knowledge graphs derived from Gene Ontology (GO) annotations and protein-protein interaction networks.

By utilizing a multi-layer Graph Neural Network (GNN), GEARS effectively propagates the localized effect of a perturbed target gene through the known network of gene-gene interactions⁵⁵. This architecture allows GEARS to accurately predict the synergistic or epistatic outcomes of perturbing novel multigene combinations, even if the constituent genes were never experimentally perturbed in the training corpus. GEARS has exhibited up to 40% higher precision than non-graph baselines in classifying complex genetic interaction subtypes, proving that explicitly mapping the biological interactome is crucial for accurate causal simulation⁵⁵.

Model	Perturbation Type	Core Architecture / Mechanism	Key Strengths in Perturbation	Limitations
scGen / CPA	Chemical & Genetic	Variational Autoencoder (VAE); Latent vector arithmetic ⁵⁶ .	Highly interpretable; excellent at interpolating seen drugs to unseen cell lines; dose-response modeling ⁵⁷ .	Fails at true extrapolation (predicting responses to entirely unseen drugs or genes) ⁵² .
GEARS	Genetic (CRISPR)	Graph Neural Network (GNN) integrated with Gene	State-of-the-art for unseen multi-gene combinatorial perturbations;	Reliant on the completeness and accuracy of external biological

		Ontology knowledge graphs ⁵⁵ .	identifies interaction subtypes (synergy, epistasis) ⁵⁵ .	knowledge graphs; limited to genetic perturbations.
PerturbNet	Chemical & Genetic	Deep Generative Model with Zero-Inflated Negative Binomial (ZINB) likelihood ⁴³ .	Directly models raw scRNA-seq counts, capturing inherent expression variability better than normalized inputs ⁴³ .	Requires extensive training data spanning diverse perturbation classes.
scGPT / Geneformer	Genetic	Massive Transformer; <i>In silico</i> masking of target genes ¹⁶ .	Leverages universal embeddings; requires no explicit external knowledge graphs; identifies general disease markers ¹⁷ .	Often fails to outperform simple linear baselines in predicting massive dynamic shifts; struggles with causal directionality ⁵ .

Foundation Model Approaches to Perturbations

Given their massive parameter counts and unparalleled exposure to diverse cellular states across billions of training tokens, single-cell foundation models are theoretically well-positioned to implicitly learn interaction manifolds without requiring hard-coded, external knowledge graphs⁵³. Models like scGPT, Geneformer, and scFoundation attempt to capture the ripple effects of genetic perturbations purely through learned self-attention weights⁹. In practice, a foundation model simulates a genetic perturbation (e.g., a CRISPR knockout) by artificially masking the target gene's token or forcing its expression value to zero at the input layer. The model then leverages its generative capacity to predict the cascading transcriptional shifts across the rest of the genome²⁶. For instance, PAH-former, a specialized model created by fine-tuning Geneformer on pulmonary arterial hypertension datasets, successfully utilized *in*

silico perturbation analysis to identify highly ranked candidate driver genes¹⁷. The simulated knockdown of the SOX18 transcription factor reversed the disease-associated transcriptional signature, a prediction subsequently validated *in vitro*, demonstrating the clinical utility of FM-based perturbation simulation¹⁷. Similarly, the generative architecture of scPRINT allows for counterfactual reasoning: by altering overarching context embeddings (such as disease state or organism identity tokens), scPRINT can synthetically generate how a mouse macrophage profile would manifest if it were expressing a human genome, yielding predicted pathway shifts that align tightly with known biology²¹.

The Extrapolation Gap and Causal Limits

Despite these highly compelling case studies, systematic, unbiased evaluations paint a heavily nuanced picture regarding the generalized efficacy of scFMs in perturbation prediction. The PertEval-scFM benchmarking framework evaluated the zero-shot embeddings of five leading scFMs against highly simplified baseline models (such as multilayer perceptrons trained directly on raw expression matrices)²⁵. The findings were stark and somewhat controversial: zero-shot scFM embeddings provided limited to no consistent improvement over simple mathematical baselines for perturbation effect prediction, particularly under severe distribution shifts²⁶. All evaluated transformer models struggled to accurately model strong, atypical, or cascading perturbation effects²⁵. This highlights a fundamental, unresolved limitation in current AI architectures: standard masked language modeling objectives—which optimize for predicting missing genes from highly correlated, simultaneous neighbors—do not naturally translate into causal, mechanistic reasoning over time⁵. While a foundation model might easily learn the statistical correlation that Gene A and Gene B are consistently co-expressed in a healthy T-cell, predicting the dynamic, non-linear transcriptomic explosion following a targeted genetic knockout requires an understanding of regulatory directionality and causality that mere statistical co-occurrence cannot provide⁵. Consequently, recent state-space architectures like RegFormer are explicitly abandoning pure self-attention in favor of baking Directed Acyclic Graphs (DAGs) of structural biology into the topological sorting of their inputs, directly addressing the gap between passive correlation and active causation³⁴.

Benchmarking and Standardized Evaluation Frameworks

The rapid, almost overwhelming influx of novel AI architectures in computational biology necessitates rigorous, standardized, and unbiased evaluation frameworks. Historically, computational models were validated on disjoint, highly curated datasets using heterogeneous metrics, obscuring true comparative performance and allowing for widespread methodological hype¹¹. Recently, comprehensive, independent benchmarks have emerged to provide holistic model assessments, strictly separating models that memorize from those that genuinely learn transferable biology.

Key Evaluation Frameworks

Benchmark Framework	Target Domain & Focus	Key Findings & Utility
CellBench-LS	Evaluates foundation models vs. classical methods in strictly low-supervision (LS) scenarios (zero-shot, few-shot) ⁶ .	FMs dominate in cell-type recognition, but classical PCA outperforms them in raw expression reconstruction tasks ⁴⁶ .
PertEval-scFM	Specifically isolates zero-shot perturbation prediction utilizing an MLP probe atop frozen embeddings ²⁶ .	Reveals that FMs fail to outperform simple baselines under strong distribution shifts, exposing the limits of current pretraining ²⁶ .
scPerturBench	Comprehensive analysis of 27 models across 29 datasets for both genetic and chemical perturbation prediction ⁶⁴ .	Highlights massive variability in model generalizability to unseen cellular contexts; promotes cellular context embedding ⁶⁴ .
SCMBench	Benchmarks 23 methods for single-cell multi-omics integration (e.g., RNA + ATAC) ⁶⁵ .	Demonstrates that fine-tuned domain-specific deep learning models frequently surpass zero-shot foundation models in multimodal alignment ⁶⁵ .

To evaluate clustering, integration, and annotation, modern benchmarks deploy a suite of stringent metrics, including the Adjusted Rand Index (ARI), Normalized Mutual Information (NMI), and the Average Silhouette Width (ASW)⁶. ASW is particularly critical for evaluating the success of batch correction. In a perfectly integrated dataset, the ASW calculated across biological cell types should approach 1 (indicating tight, well-separated biological clusters), while the ASW calculated across technical batch labels should approach 0 (indicating perfect intermixing and eradication of technical artifacts)⁶.

For the highly complex task of perturbation prediction, evaluating performance across the entire transcriptome artificially inflates performance scores, since the vast majority of the 20,000+ genes remain completely static under a localized perturbation⁴³. Therefore, the standard metric is the Mean Squared Error (MSE) computed exclusively over the top 20 or top

50 most highly variable Differentially Expressed Genes (DEGs)⁴³. Furthermore, novel evaluation metrics such as the "bio conservation score" and *scGraph-OntoRWR* have been recently introduced to quantitatively ascertain whether a model's latent space explicitly preserves known ontological relationships and biological knowledge, ensuring the model is learning true biology rather than arbitrary mathematical patterns¹⁰.

Challenges and Prospects

The transition toward foundational paradigms in single-cell biology is not without significant theoretical bottlenecks and practical roadblocks. The tension between vast parameter capacity and genuine biological fidelity drives the ongoing evolution of the field.

The Baseline Paradox and the Illusion of Complexity

A recurring and highly disruptive theme across multiple comprehensive benchmarks is the "baseline paradox": simple, classical statistical methods—such as PCA, linear regression, and domain-specific tools like scVI and Harmony—frequently remain highly competitive with, or occasionally drastically outperform, deep foundation models⁵. In the CellBench-LS evaluations, simple PCA explicitly outperformed massive foundation models in raw gene expression reconstruction tasks⁴⁶. Similarly, in real-world data integration tasks utilizing highly variable clinical samples, scVI (a targeted variational autoencoder framework) achieved the highest overall Leximax ranking for batch correction and biological structure conservation, surpassing both zero-shot and fine-tuned configurations of scGPT, scFoundation, and CellFM⁶³. This paradox suggests that the single-cell transcriptome, while superficially high-dimensional, actually occupies a relatively low-rank manifold where linear or simply parameterized probabilistic models are highly effective⁵². To justify their immense computational and environmental cost, foundation models must definitively prove their utility in regions where classical models completely fail—specifically in deep out-of-distribution reasoning, cross-species mapping, unseen perturbation extrapolation, and true multimodal inference⁵.

Optimal Feature Extraction: Layer Dynamics and the Final-Layer Fallacy

A prevailing and largely unquestioned assumption in utilizing scFMs has been that the final layer of the neural network invariably contains the most mature, holistic, and task-agnostic representation of the cell⁶⁹. However, recent systematic layer-wise representation analyses demonstrate that optimal embeddings are highly context- and task-dependent, mirroring findings in natural language processing regarding the "information bottleneck"⁶⁹.

Detailed analysis of Tahoe-X1 (a 1.3 billion parameter, 24-layer model) and scFoundation revealed that for tasks demanding the preservation of continuous biological processes (such as trajectory inference), intermediate model layers located at approximately 60% depth (e.g., layer 19 in Tahoe-X1) significantly outperformed final layers, yielding up to a 31% improvement⁶⁹. These intermediate layers capture continuous semantic transitions before the model over-optimizes for its terminal pretraining objective. Conversely, for perturbation modeling in highly activated cell states, the optimal layer shifts dynamically, while in quiescent

(inactive) cells, first-layer embeddings occasionally outperform all deeper representations⁶⁹. This paradigm-shifting discovery necessitates moving away from default final-layer extraction toward systematic layer evaluation tailored to specific biological contexts⁶⁹.

Expanding to Multimodal and Spatial Contexts

Current foundation models predominantly treat cells as isolated, independent transcriptomic bags⁷⁰. However, *in vivo* cellular phenotypes are heavily dictated by their spatial microenvironment, cell-cell communication networks, and localized epigenomic states⁵. The next major frontier in computational biology involves extending foundation models out of the suspension-cell domain and into spatial transcriptomics and tissue architecture. Models like Nicheformer and TRAILBLAZER are pioneering this critical shift by integrating spatial coordinates, local niche environments, and multicellular topologies directly into the attention mechanism⁵¹. By mathematically modeling entire "cellular neighborhoods" as the foundational unit rather than isolated cells, these architectures aim to predict multicellular, patient-level responses to pharmacological interventions, effectively bridging the gap between single-cell mechanisms and tissue-level pathology⁷⁰. Furthermore, resolving the quadratic attention bottleneck via State-Space Models (e.g., scMamba) provides the computational headroom necessary to shift from single-omic (RNA) to true pan-omic foundation models, simultaneously embedding DNA methylation, chromatin accessibility (ATAC-seq), and surface protein abundance without catastrophic memory failure, driving toward a truly holistic virtual cell³¹.

Conclusion

The integration of large-scale, self-supervised foundation model architectures into single-cell biology has undeniably accelerated the capacity to ingest, harmonize, and interpret the massive, exponential influx of transcriptomic data. Single-cell foundation models have demonstrated remarkable utility in generating unified biological latent spaces, enabling cross-species label transfer, and automating complex cell type annotations with zero-shot proficiency. However, the application of these models to complex causal inference tasks—specifically the extrapolation of transcriptional responses to novel genetic and chemical perturbations—reveals fundamental limitations in current masked-language pretraining paradigms. As evidenced by multiple independent, large-scale benchmarks, the reliance on statistical co-expression mapping does not effortlessly translate into a mechanistic understanding of biological networks, allowing classical graph-based and probabilistic baselines to maintain a competitive edge in rigorous evaluations.

The path forward for the field requires algorithmic innovation that goes beyond merely scaling parameter counts or ingesting larger datasets. The computational biology community is actively pivoting toward linear-time state-space models for full-genome inclusion, biologically grounded tokenization schemes that leverage protein structures, and multicellular spatial architectures that contextualize cells within their tissue microenvironment. By explicitly incorporating structural priors, regulatory hierarchies, and multimodal constraints into the training objectives, the next generation of foundation models will transition from sophisticated

transcriptomic pattern recognizers to true predictive *in silico* virtual cells, fundamentally transforming the landscape of computational biology, mechanistic discovery, and targeted precision therapeutics.

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